```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.read_csv("Customer Churn.csv")

data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

#replacing null value with 0 as tenure is 0 and no total charges are recorded

```python
 data["TotalCharges"] = data["TotalCharges"].replace(" ","0")
 data["TotalCharges"] = data["TotalCharges"].astype(float)

data.duplicated().sum()
```

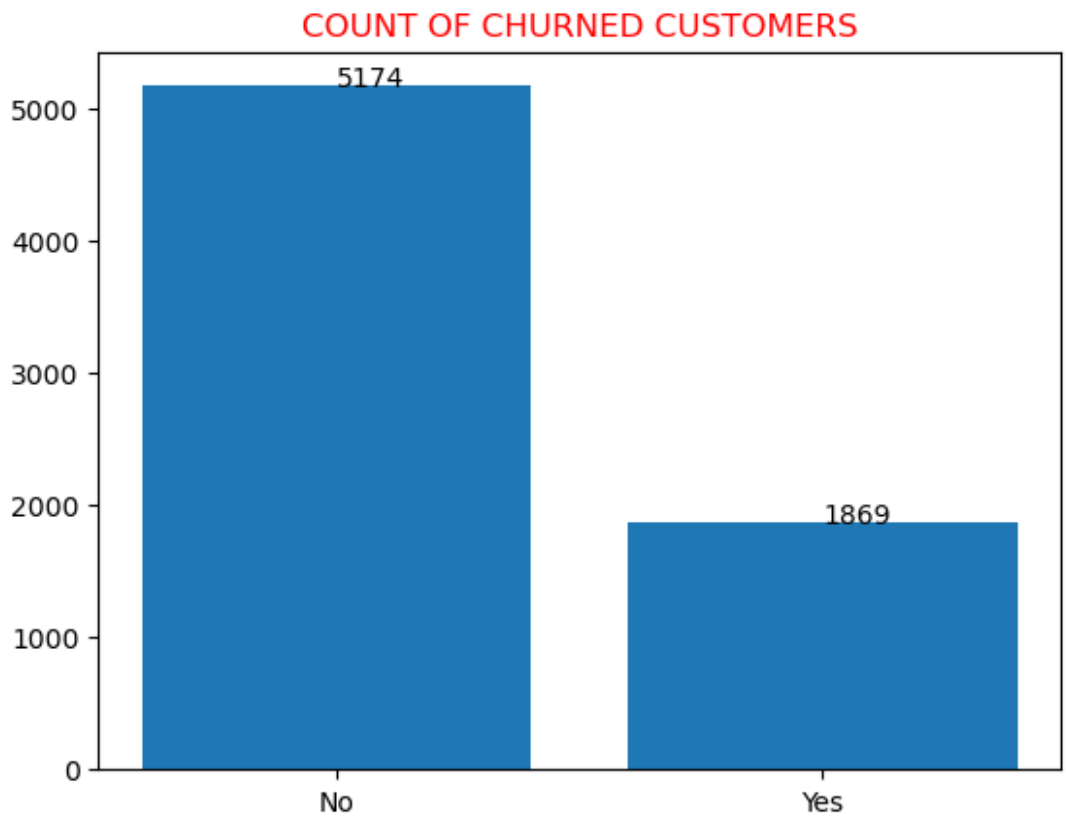
```
0
```

```python
def conv(value):
    if value == 1:
        return "YES"
```

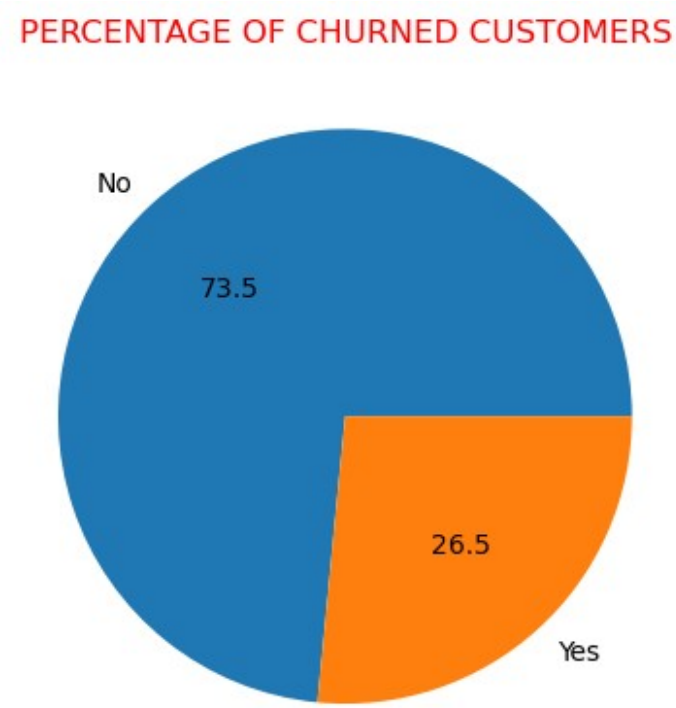
```
    else:
        return "NO"

data["SeniorCitizen"] = data["SeniorCitizen"].apply(conv)
```

CONVERTED 0 AND 1 OF SENIOR CITIZEN TO YES OR NO TO MAKE IT EASIER TO UNDERSTAND

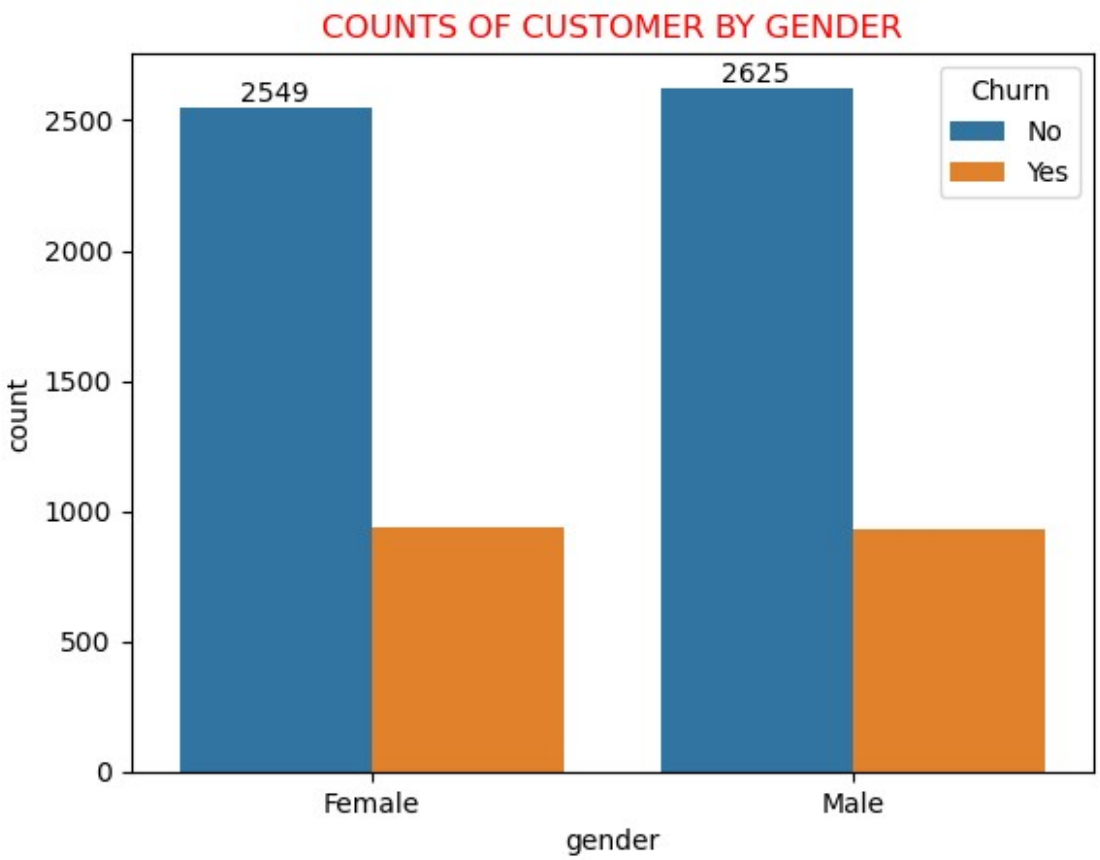
```
a = data["Churn"].value_counts()

ind = a.index
val = a.values

plt.bar(x = ind , height = val)
for i in range(0,len(ind)):
    plt.annotate(xy = (i,val[i]) , text = val[i])
plt.title("COUNT OF CHURNED CUSTOMERS" , color = "red")
plt.show()
```



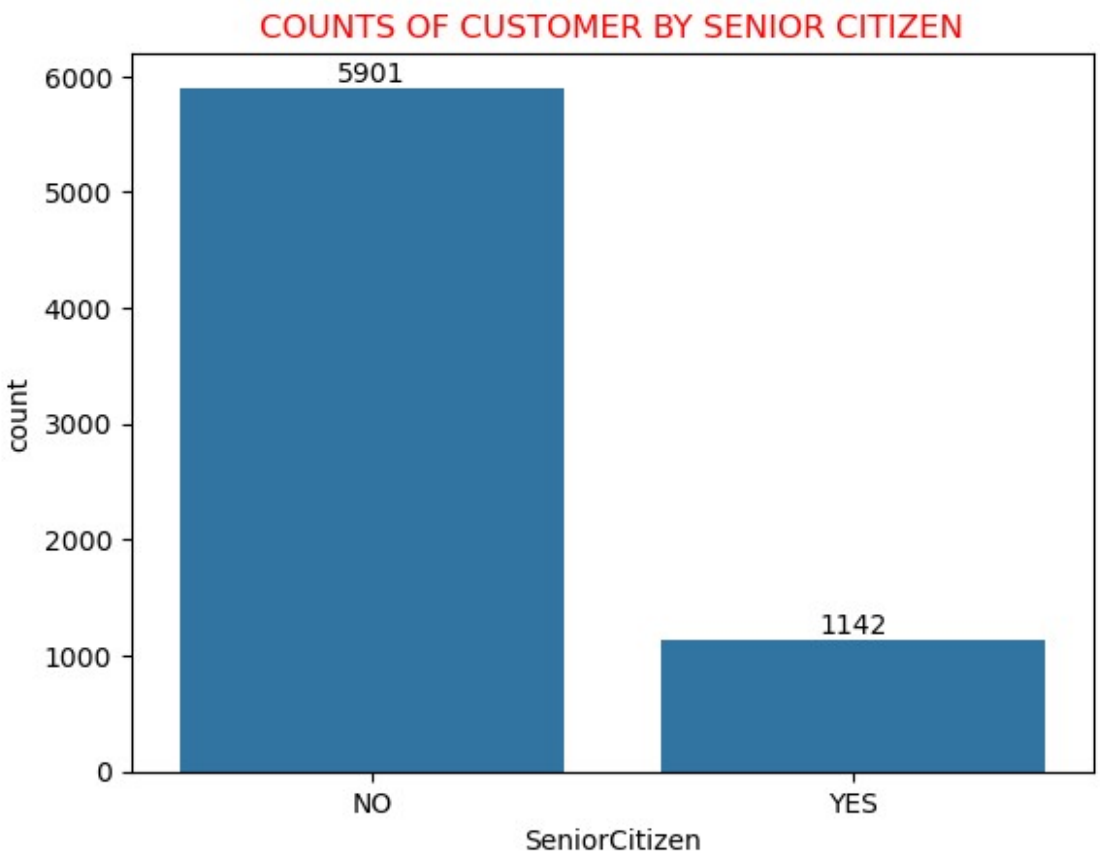
```
plt.pie(x = val , labels = ind , autopct = "%1.1f")
plt.title("PERCENTAGE OF CHURNED CUSTOMERS" , color = "red")
plt.show()
```



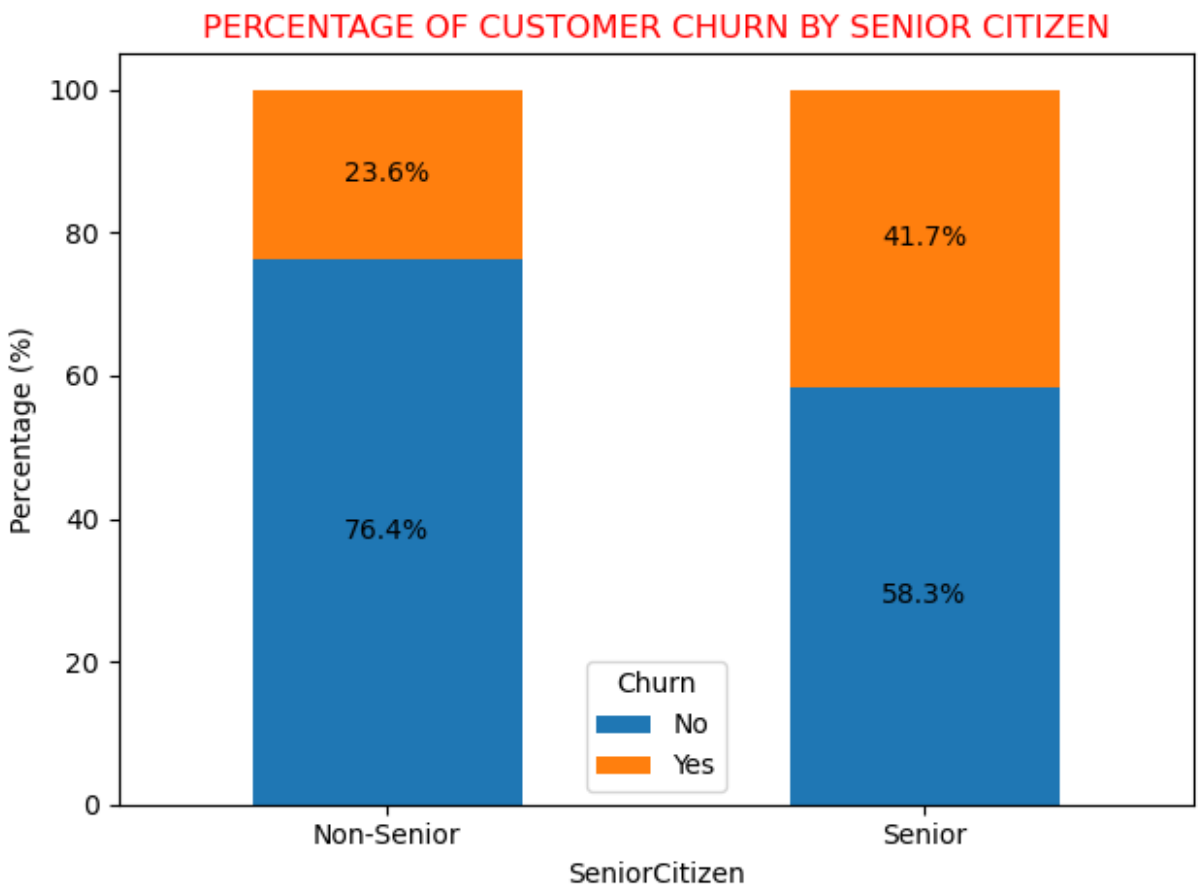

```
# from the above pie chart we can conclude that 26.5% of our customer
has been churned out

ax = sns.countplot(x = "gender" , data = data , hue = "Churn")
ax.bar_label(ax.containers[0])
plt.title("COUNTS OF CUSTOMER BY GENDER" , color = "red")
plt.show()
```




```
ax = sns.countplot(x = "SeniorCitizen" , data = data )
ax.bar_label(ax.containers[0])
plt.title("COUNTS OF CUSTOMER BY SENIOR CITIZEN" , color = "red")
plt.show()
```

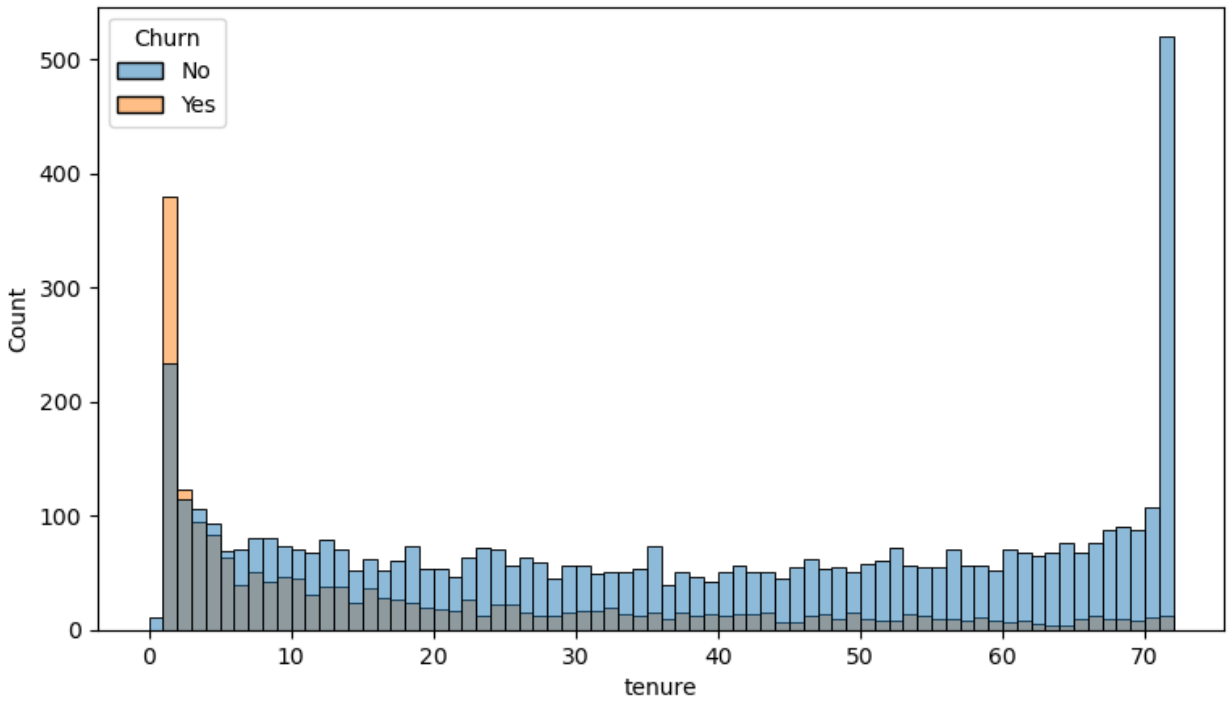



```
# Convert to percentage stacked data
stack_data = pd.crosstab(data["SeniorCitizen"], data["Churn"],
normalize="index") * 100

# Plot stacked bar chart
ax = stack_data.plot(kind="bar", stacked=True)

# Add percentage labels inside bars
for container in ax.containers:
    ax.bar_label(container, fmt="%.1f%%", label_type="center")

plt.title("PERCENTAGE OF CUSTOMER CHURN BY SENIOR CITIZEN",
color="red")
plt.xlabel("SeniorCitizen")
plt.ylabel("Percentage (%)")

# Optional: better labels instead of 0/1
plt.xticks(ticks=[0, 1], labels=["Non-Senior", "Senior"], rotation=0)

plt.legend(title="Churn")
plt.tight_layout()
plt.show()
```



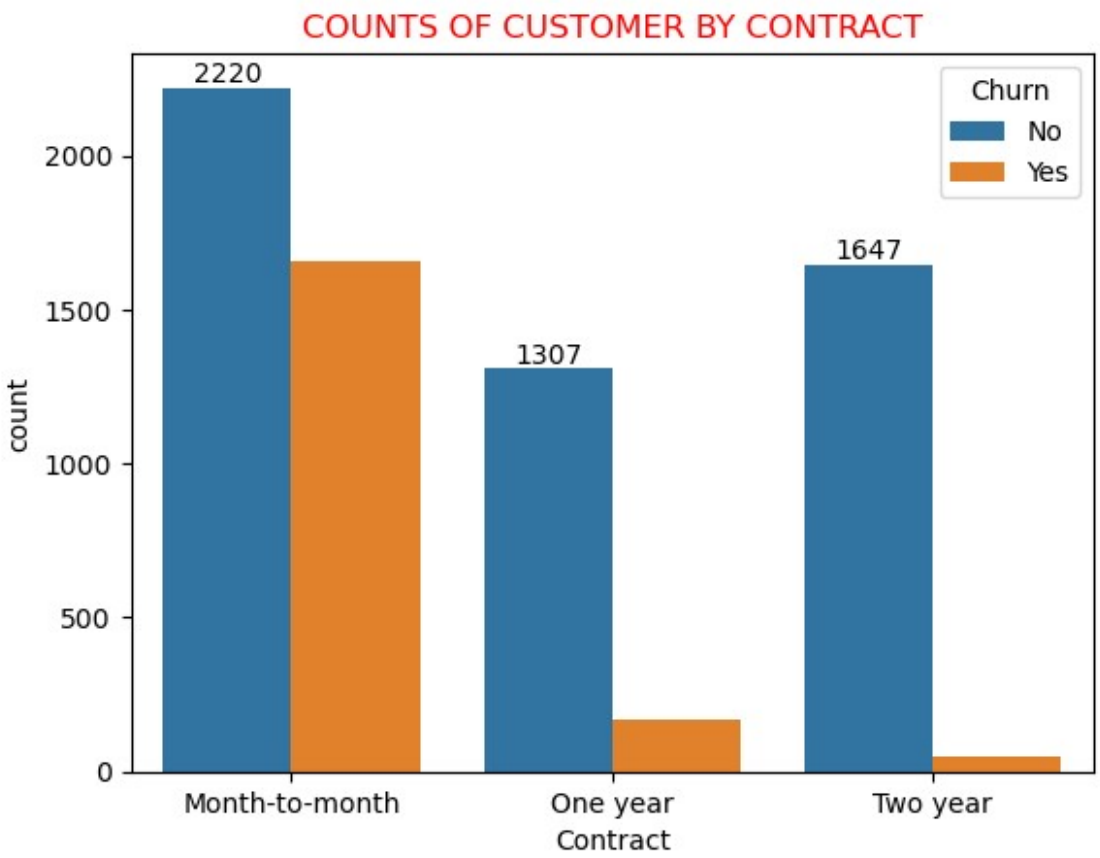
#comparitive a greater percentage in senior citizen has been churned

```
plt.figure(figsize = (9,5))
sns.histplot(x = "tenure" , data = data, bins = 72, hue = "Churn")
plt.show()
```



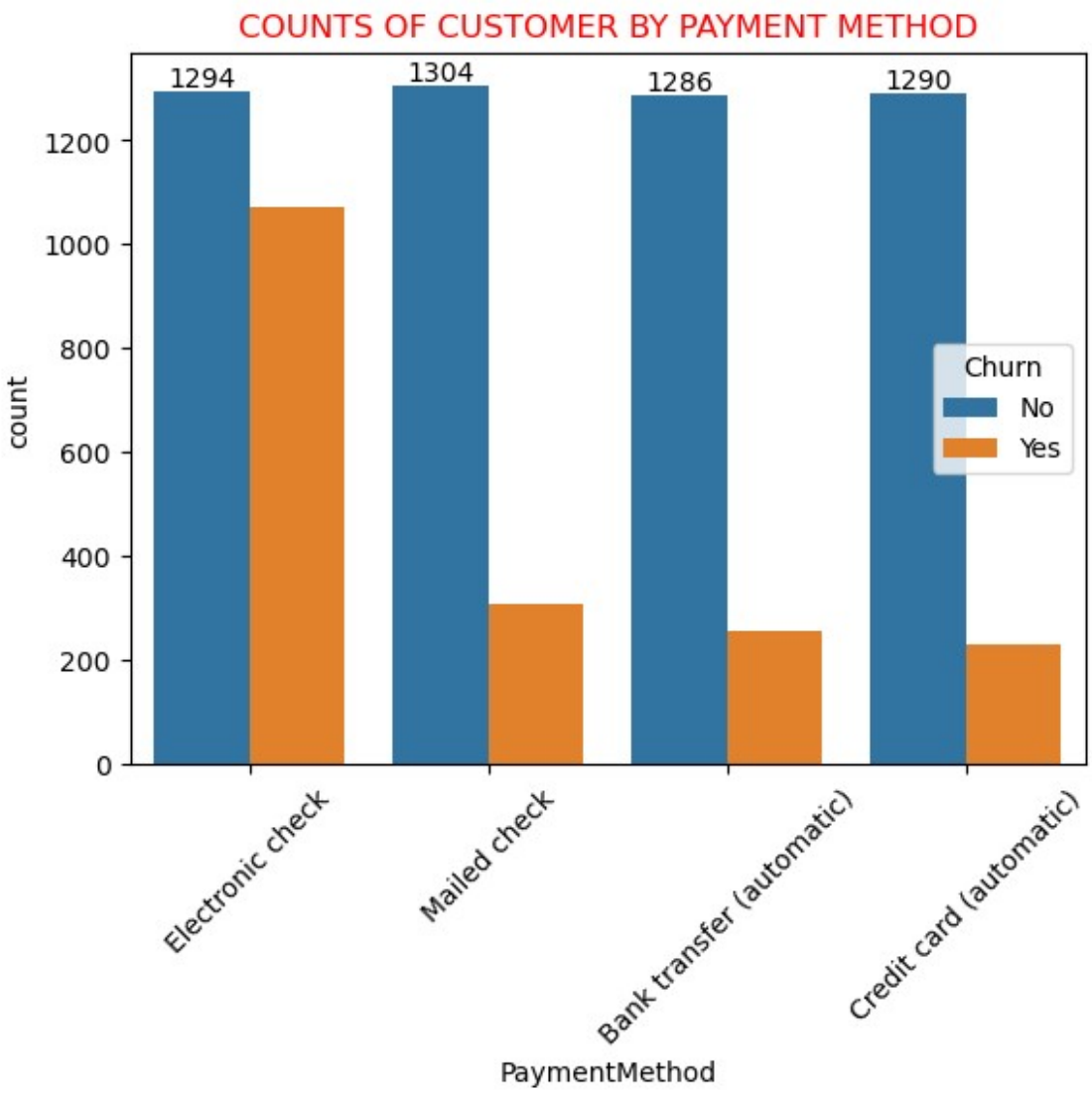
#people who had used our service for longer time have stayed and people who had used our service for 1 or 2 month had churned

```
ax = sns.countplot(x = "Contract" , data = data , hue = "Churn")
ax.bar_label(ax.containers[0])
plt.title("COUNTS OF CUSTOMER BY CONTRACT" , color = "red")
plt.show()
```



#people who have month to month contract has likely to churn than those who have 1 or 2 year contract

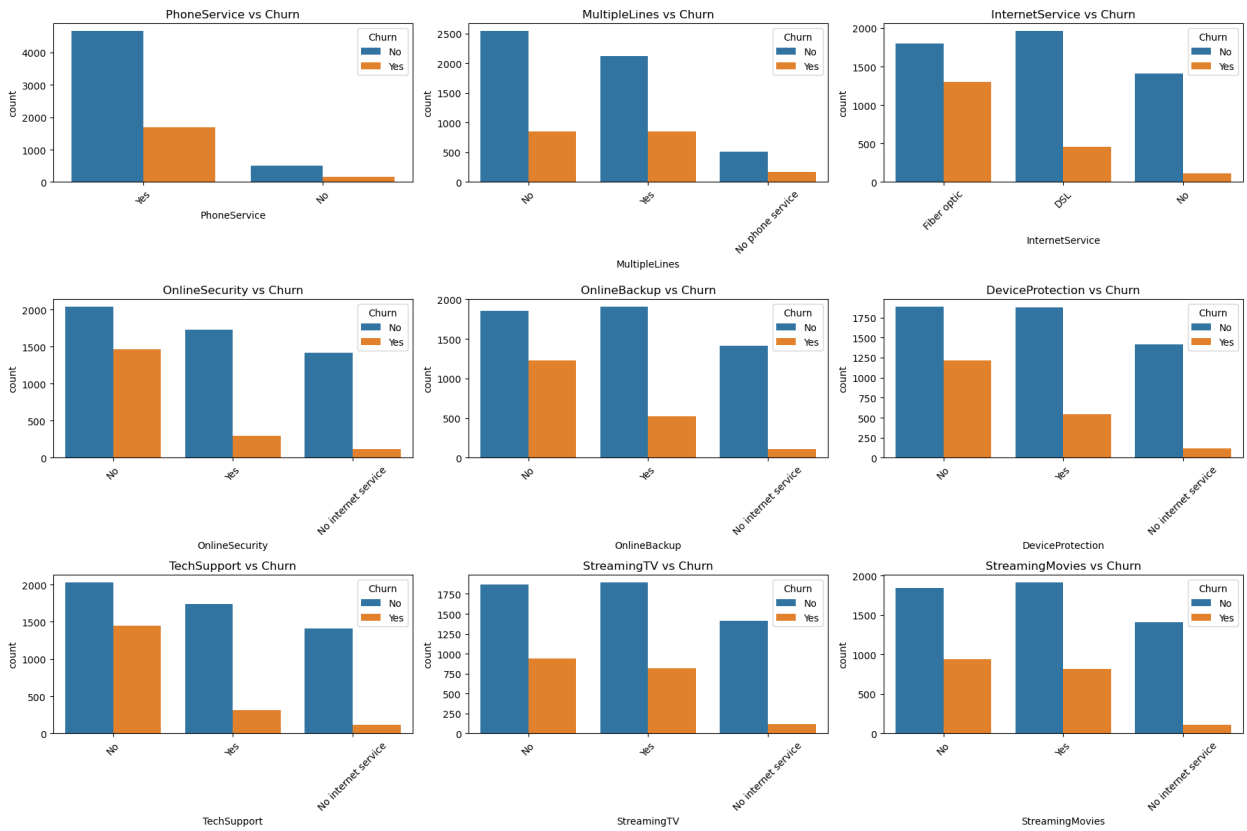
```
data.columns.values

array(['customerID', 'gender', 'SeniorCitizen', 'Partner',
'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
       'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
       'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',
       'TotalCharges', 'Churn'], dtype=object)

ax = sns.countplot(x = "PaymentMethod" , data = data, hue = "Churn")
ax.bar_label(ax.containers[0])
plt.title("COUNTS OF CUSTOMER BY PAYMENT METHOD" , color = "red")
plt.xticks(rotation = 45, ha = "center")
plt.show()
```



#CUSTOMER IS LIKELY TO CHURN WHEN HE IS USING ELECTRONIC CHECK AS A PAYMENT METHOD

```
import matplotlib.pyplot as plt
import seaborn as sns

cols = [
    'PhoneService', 'MultipleLines', 'InternetService',
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
    'TechSupport', 'StreamingTV', 'StreamingMovies'
]

plt.figure(figsize=(18, 12))

for i, col in enumerate(cols, 1):
```

```
    plt.subplot(3, 3, i)
    sns.countplot(
        x=data[col],
        hue="Churn",
        data=data,
        order=data[col].value_counts().index, # No, Yes
    )
    plt.title(f"{col} vs Churn")
    plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



#Fiber Optic Risk: Fiber optic customers exhibit a disproportionately high churn rate compared to DSL or no-internet users, marking them as the highest-risk segment.

#Retention Through Security/Support: The presence of Online Security and Tech Support significantly lowers churn. Customers without these services are much more likely to leave.

#Utility vs. Entertainment: Basic utility add-ons (Security, Backup, Device Protection) appear to be stronger retention "anchors" than entertainment add-ons (Streaming TV/Movies).

#Low-Engagement Stability: The "No Internet Service" group is the most stable, showing the lowest churn across all categories.

#Neutral Factors: Phone service and Multiple Lines have a negligible impact on the churn-to-retention ratio.